

Prove that each of the following problems is NP-hard.

1. Recall that a proper k -coloring of a graph G is a function that assigns each vertex of G a “color” from the set $\{0, 1, 2, \dots, k-1\}$ (or less formally, from any set of size k), such that for any edge uv , vertices u and v are assigned different “colors”. The *chromatic number* of G is the smallest integer k such that G has a proper k -coloring.

A proper k -coloring of a graph G is **balanced** if each color is assigned to exactly the same number of vertices. Prove that it is NP-hard to decide whether a given graph G has a *balanced* 3-coloring. [Hint: Reduce from the standard 3COLOR problem.]

2. Given an undirected graph G , does G contain a simple path that visits all but 374 vertices?
3. Given an undirected graph G , does G have a spanning tree in which every vertex has degree at most 374?
4. Given an undirected graph G , does G have a spanning tree with at most 374 leaves?

[Hint: For problems 2-4, consider the corresponding problems with 1 or 2 in place of 374.]